



MARWADI UNIVERSITY

Faculty of **Engineering and Technology**

Department of Information and Communication Technology

Enroll. No. _____

B.Tech

SEM:5

MID-SEM. EXAM: I

October: 2024

Subject: - Digital Signal and Image Processing (01CT1513)

Date:-21-10-24

Total Marks:-30

Time: - 75 Minutes

Question: 1. Explain following Terms and their uses: **CO1** [6]

- a) Impulse response b) Analog versus Digital System c) System Function d) Linear phase
e) Stable System f) Phase Delay and Group Delay

Question 2. CO2

- (a) Find the circular convolution using DFT and IDFT between {1,2,3,0} and {1,0,1,0} [6]
(b) Find 8 point DFT of following signals (a) {1,1,1,0,1,0} (b) 2^n [6]

OR

- (b) Find 8 point DIT FFT of {1,0,1,0,2,2,1,0}. [6]

Question 3. CO5

- (a) Design Analog Butterworth Filter

$$0.5 \leq |H(e^{j\omega})| \leq 1 \text{ for } 0 \leq \omega \leq \pi/2$$

$$|H(e^{j\omega})| \leq 0.2 \text{ for } 0.75\pi \leq \omega \leq \pi$$

With $T=1s$ Apply impulse invariance transformation to convert it into digital filter. [8]

- (b) Differentiate FIR and IIR Filter. [4]

OR

- (a) Explain features and architecture of Digital Signal Processor. [8]
(b) The desired response of a Low pass filter is [4]

$$H_d(e^{j\omega}) = e^{-3j\omega}, -0.75\pi \leq \omega \leq 0.75\pi$$

$$= 0; \text{ otherwise}$$

Determine FIR filter system function $H(e^{j\omega})$ for $M=3$ using a Hamming Window.

Best of Luck



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- a) Impulse response b) Analog versus Digital System c) System Function d) Linear phase
e) Stable System f) Phase Delay and Group Delay

Question 2.

- (c) Find the circular convolution using DFT and IDFT between {1,2,3,0} and {1,0,1,0} [6]
(d) Find 8 point DFT of following signals (a) {1,1,1,0,1,0} (b) 2^n [6]

OR

- (b) Find 8 point DIT FFT of {1,0,1,0,2,2,1,0}. [6]

Question 3.

- (b) Design Analog Butterworth Filter

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